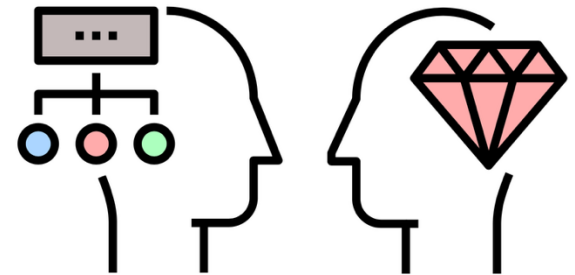


Machine Learning for Materials

3. Materials Data

Aron Walsh

Department of Materials
Centre for Processable Electronics



Module Contents

1. Introduction
 2. Machine Learning Basics
 - 3. Materials Data**
 4. Crystal Representations
 5. Classical Learning
 6. Deep Learning
 7. Building a Model from Scratch
 8. Accelerated Discovery
 9. Generative Artificial Intelligence
 10. Future Directions
-

Key Concept #3

Good research data is FAIR



Findable

persistent
identifiers



Accessible

licenses
and interface



Interoperable

machine readable
open formats

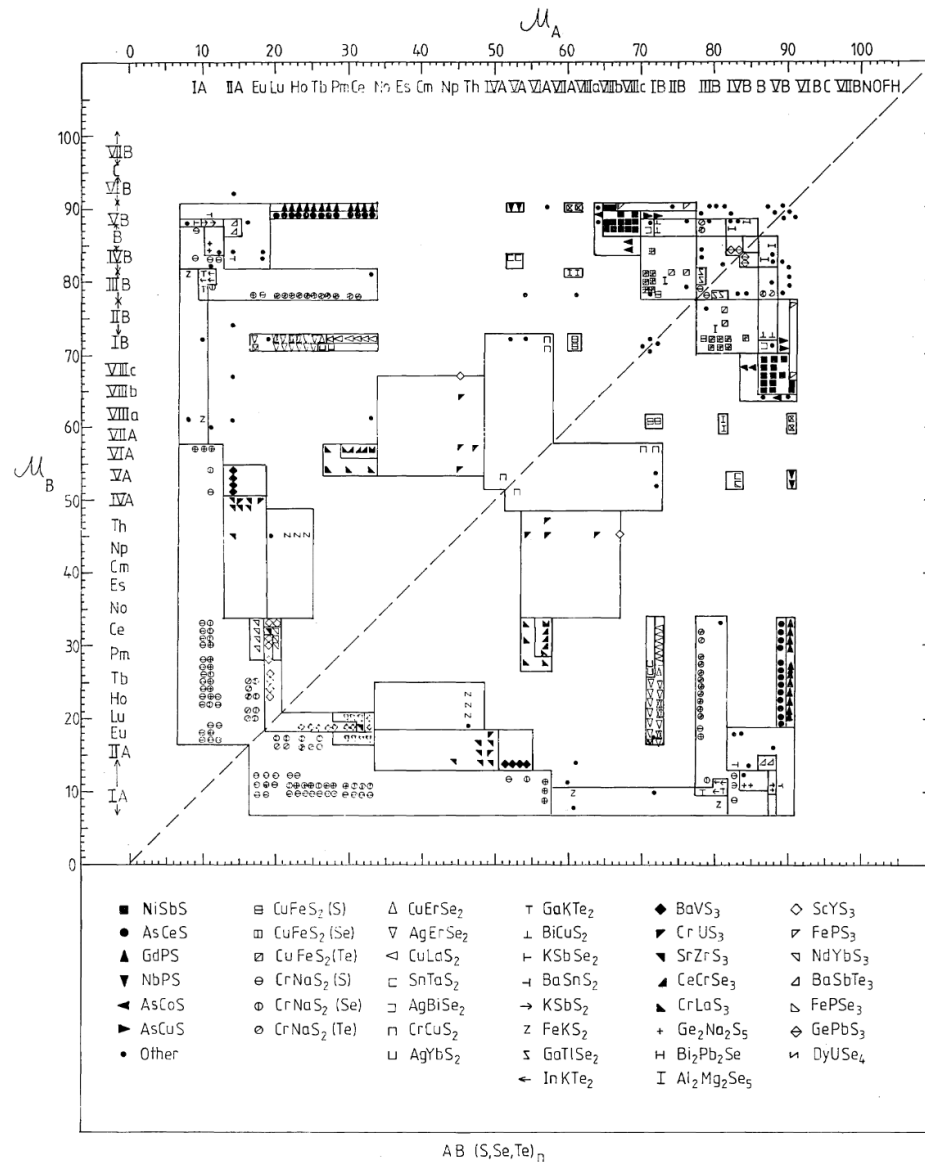


Reusable

provenance
versioning

Record the database, version, and query used to obtain your datasets

Data-Driven Materials Research



Pettifor maps
Structural classification
of ordered compounds
and alloys

Quickly predict the
structure type of new
compositions

Data-Driven Materials Research

The periodic table displays elements from Hydrogen (1) to Oganesson (118). The groups are labeled as follows:

- IA: Alkali metals
- IIA: Alkaline earth metals
- IIIB: Boron group
- IVB: Carbon group
- VB: Nitrogen group
- VIB: Oxygen group
- VIIB: Halogens
- VIIIB: Noble gases
- IIB: Zinc group
- IB: Copper group
- I: Hydrogen
- II: Helium
- III: Lithium
- IV: Beryllium
- V: Boron
- VI: Carbon
- VII: Nitrogen
- VIII: Oxygen
- IX: Fluorine
- X: Neon
- XI: Sodium
- XII: Magnesium

The lanthanide and actinide series are shown at the bottom, with elements numbered 17 through 118.

Hand-built features

- Mendeleev number for efficient grouping of structure types (periodic trends)

Data-Driven Materials Research

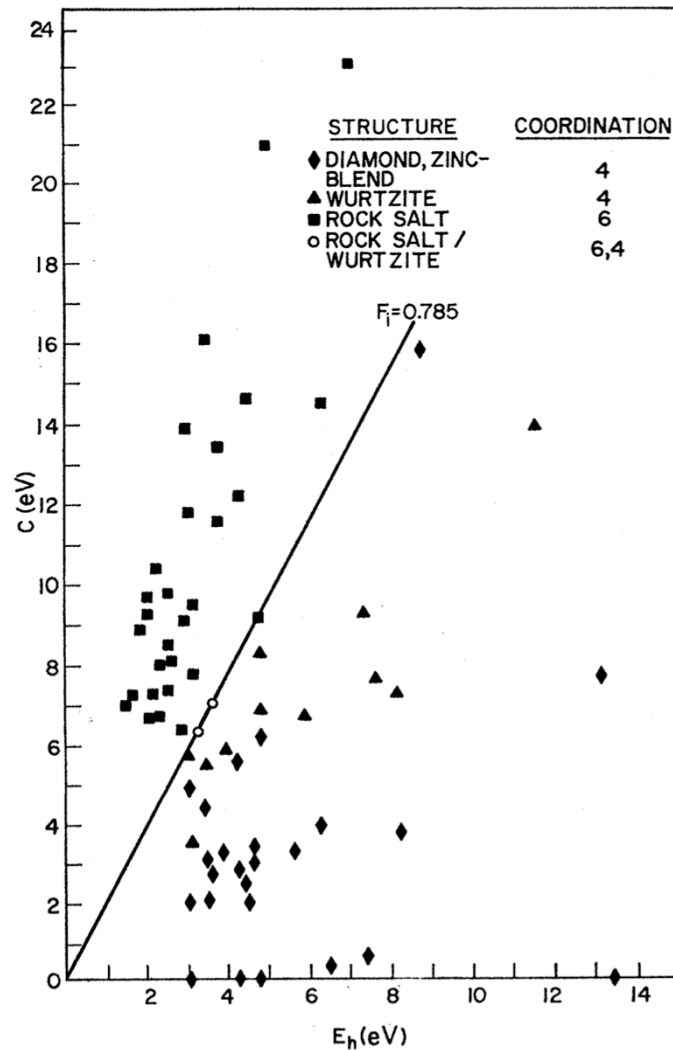


FIG. 10. The separation of fourfold and sixfold coordinates structures, using the spectroscopically defined covalent and ionic energy gaps E_h and C , respectively.

Structure-property correlations

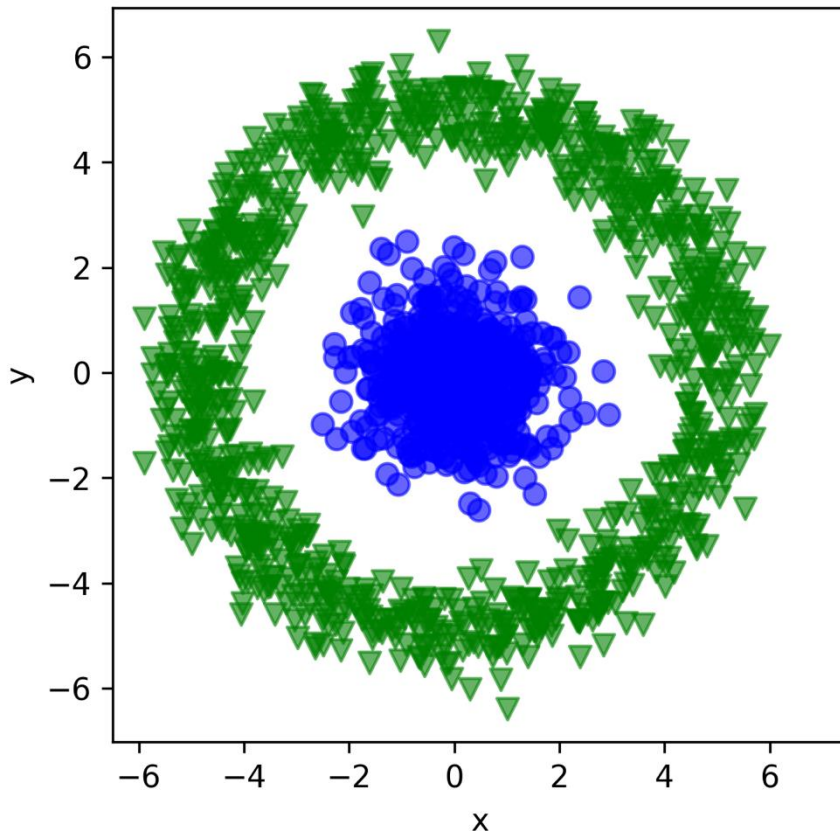
Connect crystal structure to measurable properties (mechanical, electronic, etc.)

Early analysis was manual and often focused on linear relations with physics-informed features

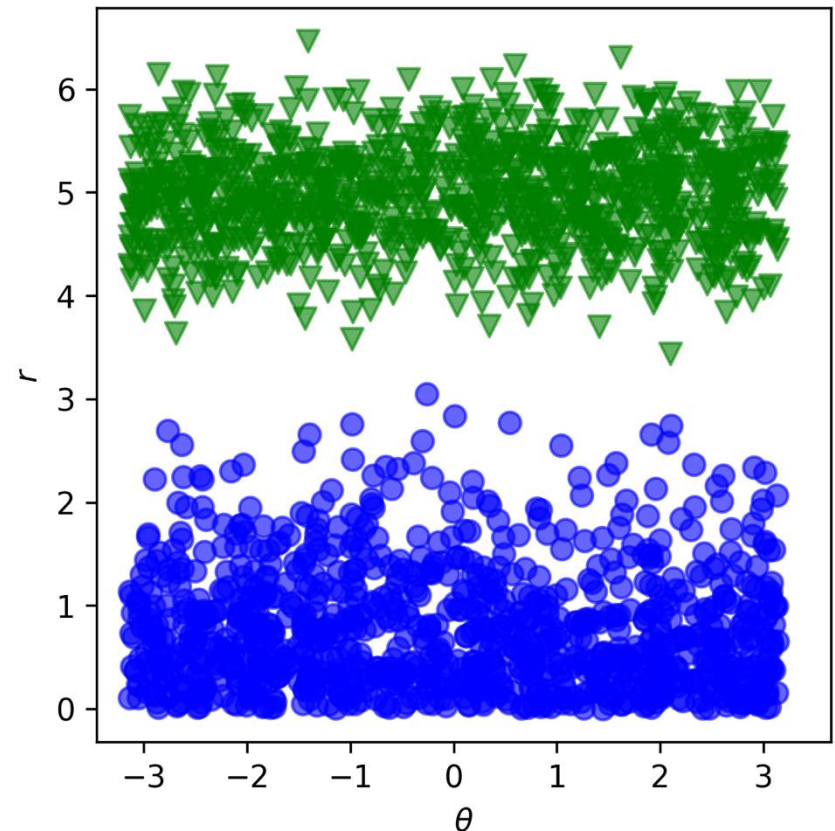
Data Representation

Choice of units or coordinate system
can greatly impact model performance

Cartesian coordinates



Polar coordinates



More on this in the next class

Class Outline

Materials Data

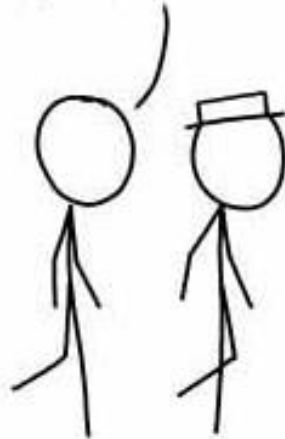
A. Data sources and formats

B. API queries

THE GREAT THING ABOUT
DIGITAL DATA IS THAT
IT NEVER DEGRADES.



HARD DRIVES FAIL,
OF COURSE, BUT THEIR
BITS CAN BE COPIED
FOREVER WITHOUT LOSS.



FILM DEGRADES, PAINT
CRACKS, BUT A COPY OF A
CENTURY-OLD DATA FILE IS
IDENTICAL TO THE ORIGINAL.

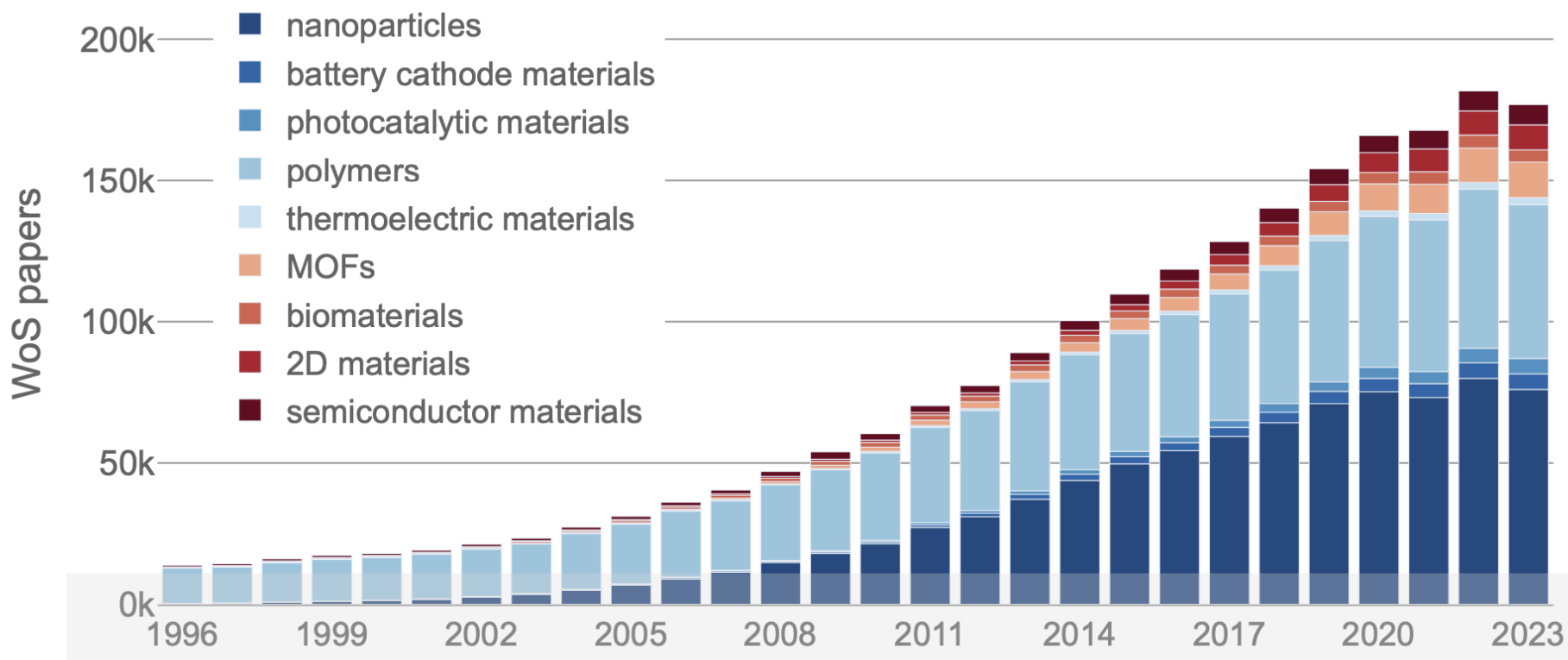


Where to Find Data?

- **Manual collection** – go through papers, extract data and tabulate (*takes time*)
- **Accelerated collection** – use of natural language processing (*requires model and workflow*)
- **Pre-built databases** – excellent when they exist in your area (*may require access fees*)
- **Automated experiments** – generate your own data over a given parameter space (*expensive*)

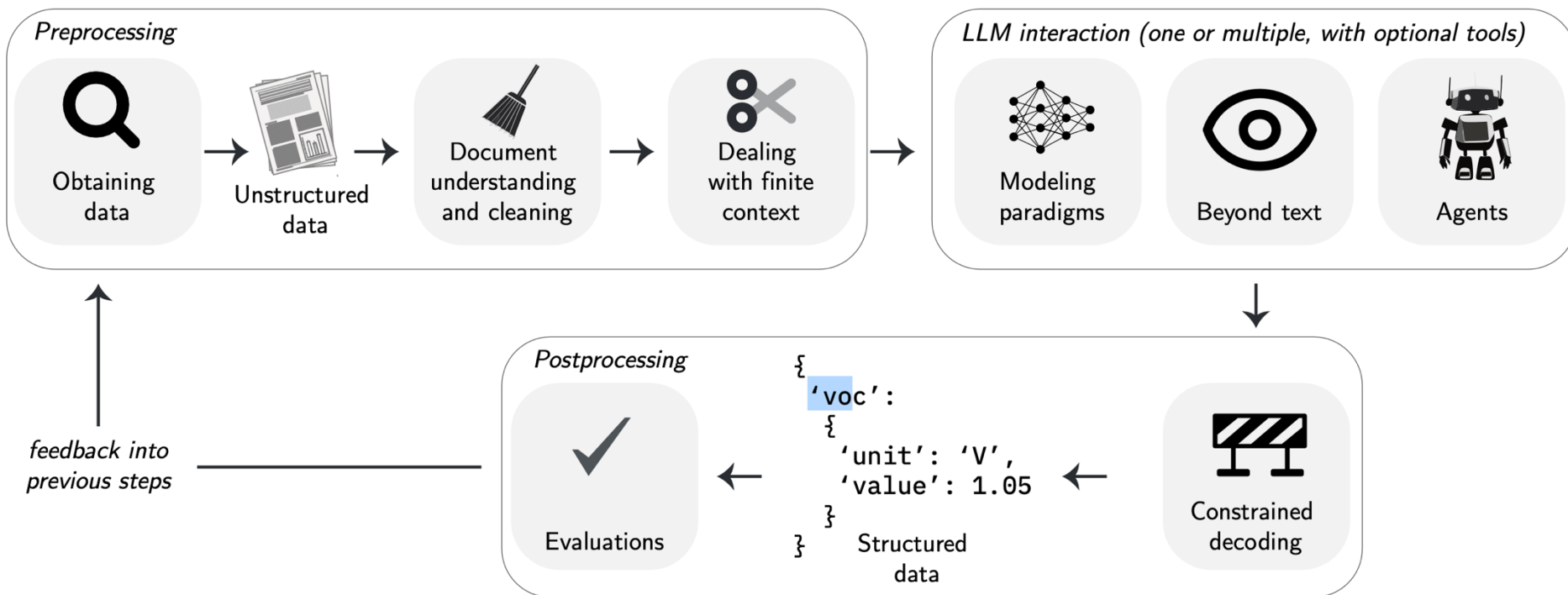
Data Extraction from the Literature

Leverage the vast literature of published papers



Data Extraction from the Literature

Many tailored workflows are available based on regular expressions and/or statistical models



Examples include <https://github.com/mcs07/ChemDataExtractor> and https://github.com/CederGroupHub/text-mined-synthesis_public

Why Share Data?

- **Reproducibility** – allow direct comparison with published literature beyond static tables and figures, e.g. raw spectra and diffraction patterns
- **Reuse** – facilitate meta-studies comparing results from multiple experiments, e.g. variation in UV-vis spectra for different samples
- **Statistical models** – power of machine learning depends on the quantity, quality, and diversity of training data

Common Forms of Data Sharing

- **Supporting information with publications** – often in the form of static pdf files (*increasingly obsolete*)
- **Data repositories** – most institutions offer data upload portals, but often lack guidelines and metadata, e.g. zip or tar files
- **Community-specific repositories** – best option if available, usually in a common format and searchable, with error detection

Common Forms of Data Sharing

Many file types that differ in how data is structured, stored, and compressed, but all easy to read in



```
import pandas as pd

# read in many structured data formats

df_csv = pd.read_csv('file.csv')

df_excel = pd.read_excel('file.xlsx')

df_hdf = pd.read_hdf('file.h5')

df_json = pd.read_json('file.json')

df_stata = pd.read_stata('file.dta')

df_clipboard = pd.read_clipboard()

df_pickle = pd.read_pickle('file.pkl')
```

JSON is common as an open, flexible, and human-readable format

FAIR Data Standards

- **Findable:** discoverable by humans & machines with metadata & persistent identifiers (e.g. DOI)
- **Accessible:** archived in long-term storage with clear access terms (e.g. CC open license)
- **Interoperable:** exchangeable between different applications and systems using open file formats
- **Reusable:** well documented and curated with clear terms and conditions on usage

Data Security

Not all databases are public, e.g. companies and academic-industrial collaborations

- **Privacy:** protection of personal data
e.g. *General Data Protection Regulation (GDPR)*
- **Encryption:** protocols for storage and transfer
e.g. *public key encryption, hashing*
- **Access control:** limiting users or computers
e.g. *passwords, firewalls*
- **Data integrity:** avoid corruption or modification
e.g. *data provenance tracking, regular versioning*

Crystallography in the Lead

Community Databases

Cambridge
Structural Database
(from 1960)

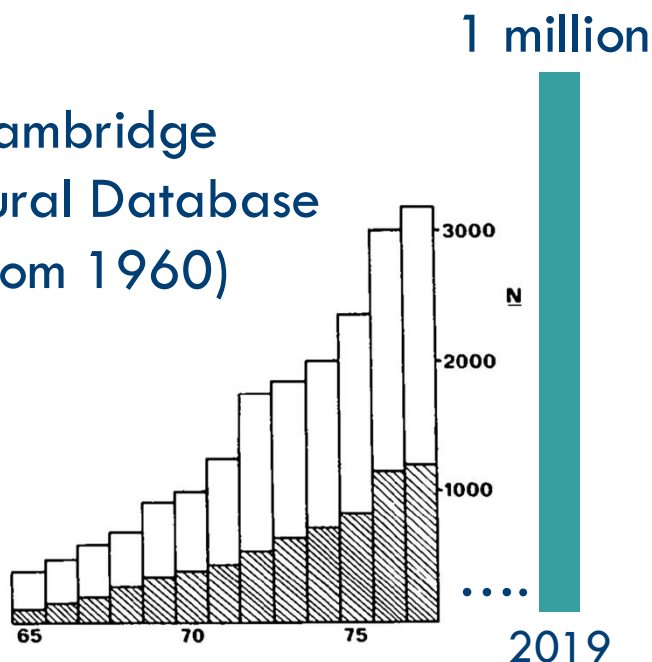


Fig. 1. Growth of CCDC database, expressed as number of entries (N) per publication year, for the period 1965–1977. The shaded area represents organometallics and metal complexes, the unshaded area represents organics.

Standard Format

Human and Machine Readable

```
1 #####
2 # Crystallographic Information Format file
3 # Produced by PyCifRW module
4 #
5 # This is a CIF file. CIF has been adopted by the International
6 # Union of Crystallography as the standard for data archiving and
7 # transmission.
8 #
9 # For information on this file format, follow the CIF links at
10 # http://www.iucr.org
11 #####
12
13 data_SnS2
14 _symmetry_space_group_name_H-M 'P 1'
15 _cell_length_a 3.6999199
16 _cell_length_b 3.6999199
17 _cell_length_c 6.97795855
18 _cell_angle_alpha 90.0
19 _cell_angle_beta 90.0
20 _cell_angle_gamma 119.999999964
21 _chemical_name_systematic 'Generated by pymatgen'
22 _symmetry_Int_Tables_number 1
23 _chemical_formula_structural SnS2
24 _chemical_formula_sum 'Sn1 S2'
25 _cell_volume 82.7263116646
26 _cell_formula_units_Z 1
27 loop_
28 _symmetry_equiv_pos_site_id
29 _symmetry_equiv_pos_as_xyz
30 1 'x, y, z'
```

Crystallography in the Lead

Compound composition ,
unit cell dimensions
and symmetry

```
data_2010|jaa0012_p21n
_chemical_formula_sum
'F3 H4 Mn N O3 P'
_chemical_formula_weight      208.94
_symmetry_cell_setting        monoclinic
_symmetry_space_group_name_H  -M  P21/n
_cell_length_a                 5.3999(2)
_cell_length_b                 13.1644(6)
_cell_length_c                 7.4117(3)
_cell_angle_alpha              90.00
_cell_angle_beta               107.501( 3)
_cell_angle_gamma              90.00
_cell_volume                   502.48(4)
_cell_formula_units_Z          4
```

```
_cell_measurement_temperature  293(2)
_diffraction_ambient_temperature 120(2)
_diffraction_radiation_wavelength 0.71073
_diffraction_radiation_type      MoK  \a
_diffraction_radiation_source     'Bruker  -Nonius FR591 rotating anode'
_diffraction_detector_area_resol_mean '4096x4096pixels / 62x62mm'
_diffraction_reflections_number   6305
_diffraction_reflections_av_R_equivalents 0.0561
_diffraction_reflections_av_sigma/netI 0.0356
_refinement_ls_R_factor_all       0.0344
_refinement_ls_R_factor_gt        0.0294
_refinement_ls_wR_factor_ref      0.0782
_refinement_ls_wR_factor_gt       0.0737
_refinement_ls_goodness_of_fit_ref 0.928
_refinement_ls_restrained_S_all   0.928
```

Data collection and
refinement information

Atom coordinates

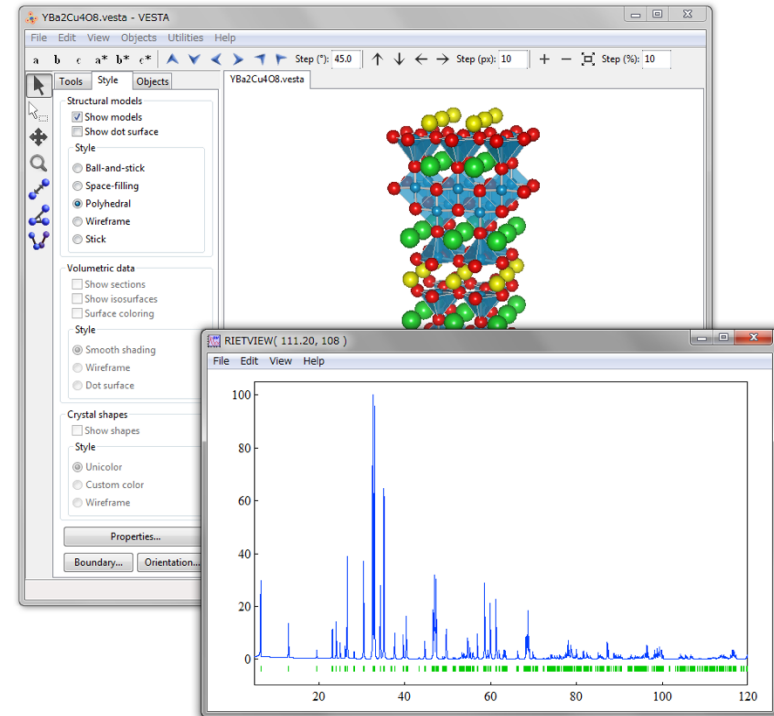
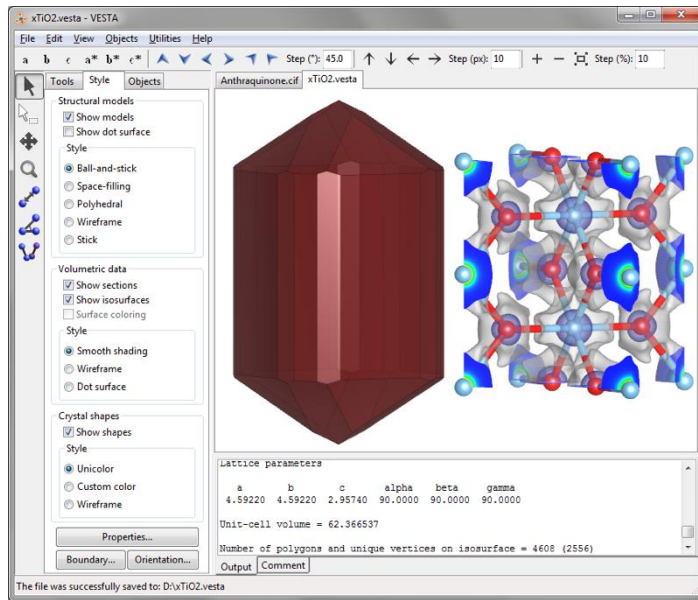
```
H4 H 0.563(8) 0 .763(4) 1.186(6) 0.036(12) Uiso 1 1 d ...
Mn1 Mn 0.0000 0.5000 0.5000 0.00741(16) Uani 1 2 d S ...
P1 P 0.44334(12) 0.39660(5) 1.31194(9) 0.00852(17) Uani 1 1 d ...
Mn2 Mn 0.0000 0.5000 1.0000 0.00813(17) Uani 1 2 d S ...
F1 F 0.4112(3) 0.28867(12) 1.2 175(2) 0.0134(3) Uani 1 1 d ...
F2 F 0.0669(3) 0.44659(11) 0.7416(2) 0.0115(3) Uani 1 1 d ...
O3 O 0.2656(3) 0.59984(13) 0.5925(2) 0.0105(4) Uani 1 1 d ...
F3 F 0.0951(3) 0.62898(12) 0.9601(2) 0.0130(3) Uani 1 1 d ...
```

```
O1 0.0097(9) 0.0128(9) 0.011 4(8) 0.0016(6) 0.0056(7) 0.0019(7)
O2 0.0088(9) 0.0152(9) 0.0098(8) 0.0022(7) 0.0025(7) 0.0001(7)
N1 0.0122(12) 0.0121(12) 0.0133(12) -0.0005(9) 0.0021(9) -0.0009(9)
Mn1 0.0068(3) 0.0091(3) 0.0062(3) 0.00054(17) 0.0017(2) -0.00003(18)
P1 0.0075(3) 0.0103(3) 0.0077(3) -0.0006(2) 0.0023(2) 0.0004(2)
Mn2 0.0075(3) 0.0096(3) 0.0067(3) 0.00087(18) 0.0012(2) -0.00011(19)
F1 0.0134(8) 0.0112(7) 0.0157(8) -0.0032(6) 0.0042(6) -0.0006(6)
F2 0.0124(7) 0.0134(7) 0.0085(7) 0.0020(6) 0.0030(6) 0.0024(6)
O3 0.0079(9) 0.0113(9) 0.0120(9) 0.0002(6) 0.0025(7) 0.0009(7)
F3 0.0132(8) 0.0128(7) 0.0115(7) 0.0019(6) 0.0014(6) -0.0014(6)
```

Atomic displacement parameters

Crystallography in the Lead

Many open-source programs for cif visualisation
(including Miller indices, diffraction patterns...)



Example: General Repository



[Communities](#)[My dashboard](#)

September 26, 2025 (v1)

Dataset

Open

Dynamic Vacancy Levels in CsPbCl₃ Obey Equilibrium Defect Thermodynamics

Mosquera-Lois, Irea ; Walsh, Aron 

Datasets and trained MACE models for V_{Cl}^0 and V_{Cl}^{+1} in CsPbCl₃, used in the publication "Dynamic Vacancy Levels in CsPbCl₃ Obey Equilibrium Defect Thermodynamics" (<https://arxiv.org/abs/2508.14513>). The data is provided as a compressed folder named "08_Data.tar.gz"...

Part of [Materials Design Group](#)

Uploaded on September 26, 2025

 68  9

July 31, 2025 (v3.2.0)

Software

Open

WMD-group/SMAC: v3.2.0

Anthony Onwuli; Alex Moriarty; Dan Davies; and 15 others

<!-- Release notes generated using configuration in .github/release.yml at master --> What's Changed 🦄 Other Changes Update PyPI publish workflow and Changelog workflow by @AntObi in <https://github.com/WMD-group/SMAC/pull/402> Combined PR by @github-actions[bot] in...

Part of [Materials Design Group](#)

Uploaded on July 31, 2025

19 more versions exist for this record

 1525  320

May 23, 2025 (v1)

Dataset

Open

Defect Tolerance via External Passivation in the Photocatalyst SrTiO₃:Al

Ogawa, Kanta 

Accompanying data for "Defect Tolerance via External Passivation in the Photocatalyst SrTiO₃:Al"

Part of [Materials Design Group](#)

Uploaded on May 29, 2025

 84  77

Example: Community Repository

PUBLISH ▾EXPLORE ▾ANALYZE ▾ABOUT ▾
Entries search 

LOGIN / REGISTER  UNITS

ENTRIES SEARCH FILTERS    

Material >

Elements / Formula >

Elements

Fe  0 

Symmetry >

Method >

Simulation >

DFT >

GW >

Experiment

EELS >

20/21,504 search results 

Example: Curated Repository

Welcome to ICSD Web. IP authenticated (130.246.140.4). Royal Society Chemistry

FIZ Karlsruhe | Contact
Close session

Detailed ViewEntry 1 of 1

[Back to Query](#)[Back to List](#)[CCDC](#)[Export Cif](#)[Print](#)[Feedback to Editor](#)

SummaryCollection Code 26170

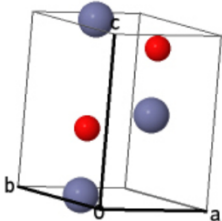
Struct. formula	Zn O	Structure type	Wurtzite#ZnS(2H)
Cell parameter	3.24986(1) 3.24986(1) 5.20662(1) 90. 90. 120.	Space group	P 63 m c (186)
Cell volume	47.62 [Å³]	Z	2
Temperature	room temperature	Pressure	atmospheric
Data quality	High quality	R-value	0.0378
Author	Abrahams, S.C.; Bernstein, J.L.	Title	Remeasurement of the structure of hexagonal Zn O
Reference	Acta Crystallographica, Section B: Structural Crystallography and Crystal Chemistry (1969) 25, p. 1233-1236	DOI	10.1107/S0567740869003876 ↗

DetailsExpand allCollapse all?

Visualization

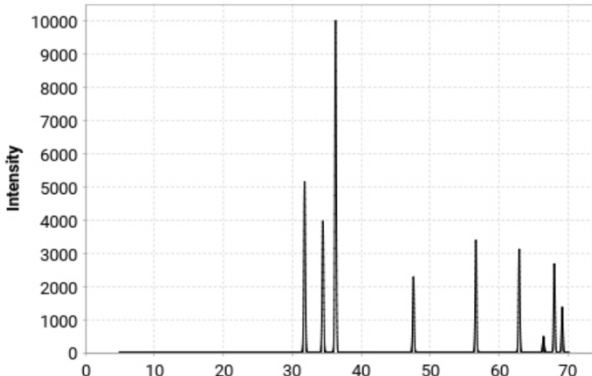
Published Crystal Structure

HM:P 63 m c #186
a=3.250Å
b=3.250Å
c=5.207Å
α=90.000°
β=90.000°
γ=120.000°



[Interactive Visualization](#)

Powder Pattern



[Interactive Visualization](#)

Example: Materials Modelling

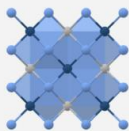


Materials Explorer

App by Materials Project

 References

 Documentation



Ac₂AgI₂

mp-861724

TABLE OF CONTENTS

Summary

Crystal Structure

Properties

Thermodynamic Stability

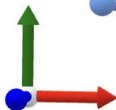
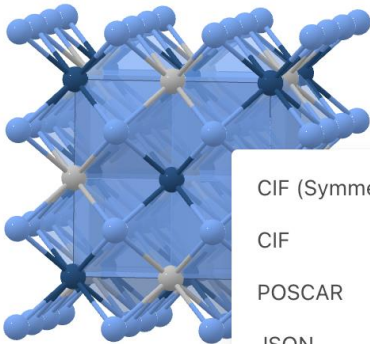
Electronic Structure

Phonon Dispersion

Diffraction Patterns

Aqueous Stability

Magnetic Properties





CIF (Symmetrized)

CIF

POSCAR

JSON

Prismatic

VASP Input Set (MPRelaxSet)

Energy Above Hull	0.000 eV/atom
Space Group	Fm $\bar{3}$ m
Band Gap	0.00 eV
Predicted Formation Energy	-0.372 eV/atom
Magnetic Ordering	Non-magnetic
Total Magnetization	0.00 μ B/f.u.
Experimentally Observed	No

Example: Microscopy



The screenshot shows the EMDB (Electron Microscopy Data Bank) website. The header features navigation links for EMBL-EBI, Services, Research, Training, and About us. The main banner includes the EMDB logo and a search bar with the text "Enter your search term(s) in the box below or [build an advanced search query](#)". Below the search bar, examples of search terms are provided: "1001, Apoferritin, Tomography, Rossmann MG, 5A1A". A secondary navigation bar contains links for Home, Deposition, Documentation, Resources, FTP Archive, REST API, About, Feedback, and Share. The main content area is titled "About EMDB" and contains a paragraph describing the database as a public repository for electron cryo-microscopy volume maps and tomograms. It mentions its founding in 2002 under Kim Henrick's leadership and its current operation under the Worldwide Protein Data Bank (wwPDB). A list of collaborating archives (EMDB, PDB, and EMPIAR) is provided. A "Quick links" sidebar on the right lists various resources like EMDB Policies, Talks & Tutorials, Validation Analysis, Volume Browser, EMDB Citations, EMPIAR, PDBe, Biolmage Archive, EMDataResource, and EM Navigator.

EMBL-EBI Services Research Training About us EMBL-EBI

EMDB
Electron Microscopy Data Bank

Enter your search term(s) in the box below or [build an advanced search query](#)

Search EMDB...

Examples: 1001, Apoferritin, Tomography, Rossmann MG, 5A1A

Home Deposition Documentation Resources FTP Archive REST API About Feedback Share

About EMDB

The Electron Microscopy Data Bank ([EMDB](#)) is a public repository for electron cryo-microscopy volume maps and tomograms of macromolecular complexes and subcellular structures. It covers a variety of techniques, including single-particle analysis, electron tomography, and electron crystallography (for more information, see the [EMDB Policies](#)).

EMDB was founded at [EMBL-EBI](#) in 2002 under the leadership of Kim Henrick. From 2007, the archive has been operated jointly by EMBL-EBI and the [Research Collaboratory for Structural Bioinformatics \(RCSB\)](#)^a. In 2013, the [Protein Data Bank Japan \(PDBj\)](#) also became involved in EMDB. As of the first of January, 2021, EMDB is an archive operated under the aegis of the [Worldwide Protein Data Bank \(wwPDB\)](#), and has also joined that organisation as a full member.

Results of cryo-EM studies are archived in three collaborating archives:

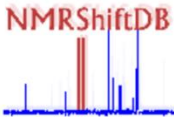
- EMDB stores the processed 3D volumes and tomograms;
- PDB stores any atomic models that have been constructed based on EMDB data;
- EMPIAR stores the raw EM data underpinning the data in EMDB/PDB.

Data for EMDB and PDB can be deposited through the wwPDB deposition and annotation system [OneDep](#).

Quick links

- [EMDB Policies](#)
- [Talks & Tutorials](#)
- [Validation Analysis](#)
- [Volume Browser](#)
- [EMDB Citations](#)
- [EMPIAR](#)
- [PDBe](#)
- [Biolmage Archive](#)
- [EMDataResource](#)
- [EM Navigator](#)

Example: NMR



Current usage is:
Registered Users: 2032
Structures: 236869
Spectra: Measured 399439, calculated 554

Username:

Password:

☐ Use cookies for persistent login

Login

[Create New Account](#)
(Only necessary for contributing data)

[Forgot password?](#)

[Impressum](#)

Problems using nmrshiftdb2? See our [tips on browsers to use](#) !

Home Search Results Quick Check Predict Assignment Submit Review Help

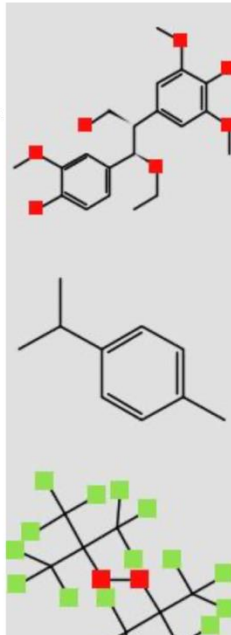
nmrshiftdb2 Links

[nmrshiftdb2 Documentation](#)
[Developers' page](#)
[Media coverage](#)
[Links](#)
[FAQ](#)
[Guestbook](#)
[Contact](#)

About nmrshiftdb2

nmrshiftdb2 is a NMR database (web database) for organic structures and their nuclear magnetic resonance (nmr) spectra. It allows for spectrum prediction (^{13}C , ^1H and other nuclei) as well as for searching spectra, structures and other properties. The nmrshiftdb2 software is open source, the data is published under an open content license. The core of nmrshiftdb2 are fully assigned spectra with raw data and peak lists (we have pure peak lists as well). Those datasets are peer reviewed by a [board of reviewers](#). The project is supported by a [scientific advisory board](#). nmrshiftdb2 is part of the [NFDI4Chem initiative](#) and will provide a component for a curated repository there. Please consult the [documentation](#) for more detailed information.

Latest Additions



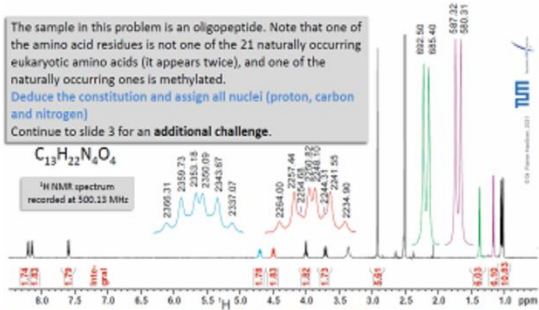
nmrshiftdb2 collaborates with



The sample in this problem is an oligopeptide. Note that one of the amino acid residues is not one of the 21 naturally occurring eukaryotic amino acids (it appears twice), and one of the naturally occurring ones is methylated. Deduce the constitution and assign all nuclei (proton, carbon and nitrogen). Continue to slide 3 for an additional challenge.

$\text{C}_{13}\text{H}_{22}\text{N}_4\text{O}_4$

^1H NMR spectrum recorded at 500.13 MHz



Which molecule is this? See our [Problem of the Month presentation](#) to find out.

Quick Check



[Check your assignment here](#)

[View Problem of the Month archive from 2020](#)
[View old Problem of the Month archive](#)

Class Outline

Materials Data

A. Data sources and formats

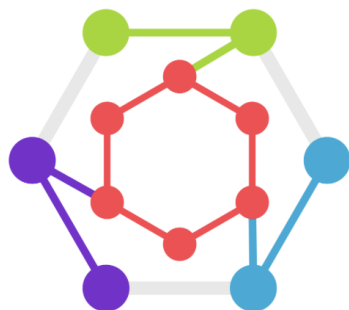
B. API queries

Database Access

Mode	Advantage	Disadvantage
Web browser	No knowledge of database software is required	Often one material at a time – slow for large datasets
Data file	All data is downloaded as one (e.g. zip or tar) file	Specialist software often needed; data is not up-to-date
API* (e.g. Python)	Access latest data with advanced queries	Some programming knowledge required

*API = Application Programming Interface

Materials Database Access: Python API



OPTIMADE

Open Databases Integration
for Materials Design

[About](#)[Resources](#)[Specification](#)[Contributors](#)[GitHub](#)[Forum](#)[Try It!](#)

About us

The **Open Databases Integration for Materials Design** (OPTIMADE) consortium aims to make materials databases interoperable by developing a specification for a common REST API.

We have released version 1 of the OPTIMADE specification, with several databases already providing implementations. A full list is available on the [OPTIMADE providers dashboard](#).

24

29

27,730,600

PROVIDERS

DATABASES

STRUCTURES

Query – Optimade



```
# Import OptimadeClient
from optimade.client import OptimadeClient

# Initialise client for OQMD provider, synchronous
client = OptimadeClient(include_providers=["oqmd"], use_async=False)

# Get structures with reduced chemical formula "OPb"
results = client.get(filter='chemical_formula_reduced="OPb"')
```

Query – Materials Project (MPRester)



```
# Import MPRester from the Materials Project API
from mp_api.client import MPRester
from emmet.core.summary import HasProps

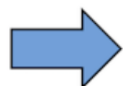
# Access the Materials Project
with MPRester("your_api_key_here") as mpr:
    # Search for materials with dielectric properties
    docs = mpr.materials.summary.search(
        has_props=[HasProps.dielectric],
        fields=["material_id"] # Only retrieve the material_id field
    )

# Extract material IDs from the search results
mpids = [doc.material_id for doc in docs]
```

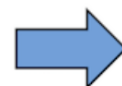

Load a Dataset



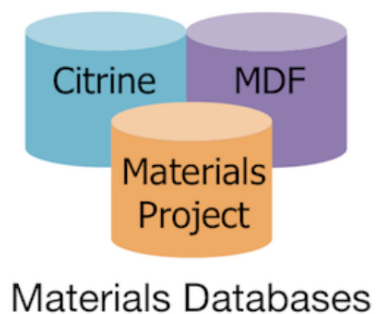
**Data
Retrieval**



**Feature
Extraction**



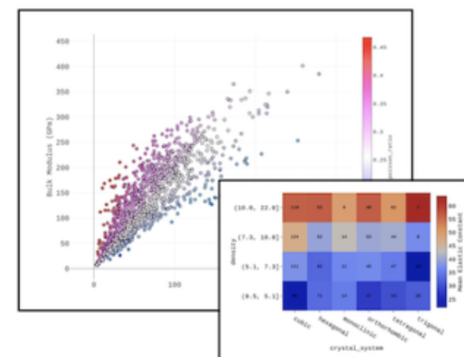
Visualization



MATERIAL	FEATURES				PROPERTY
TiO ₂ rutile	F ₁₁	F ₁₂	...	F _{1N}	gap = 3.0 eV
C diamond	F ₂₁	F ₂₂	...	F _{2N}	gap = 5.5 eV
...	...	...	...	...	...
PbTe rocksalt	F _{M1}	F _{M2}	...	F _{MN}	gap = 0.3 eV



**Machine
Learning**



Load a Dataset

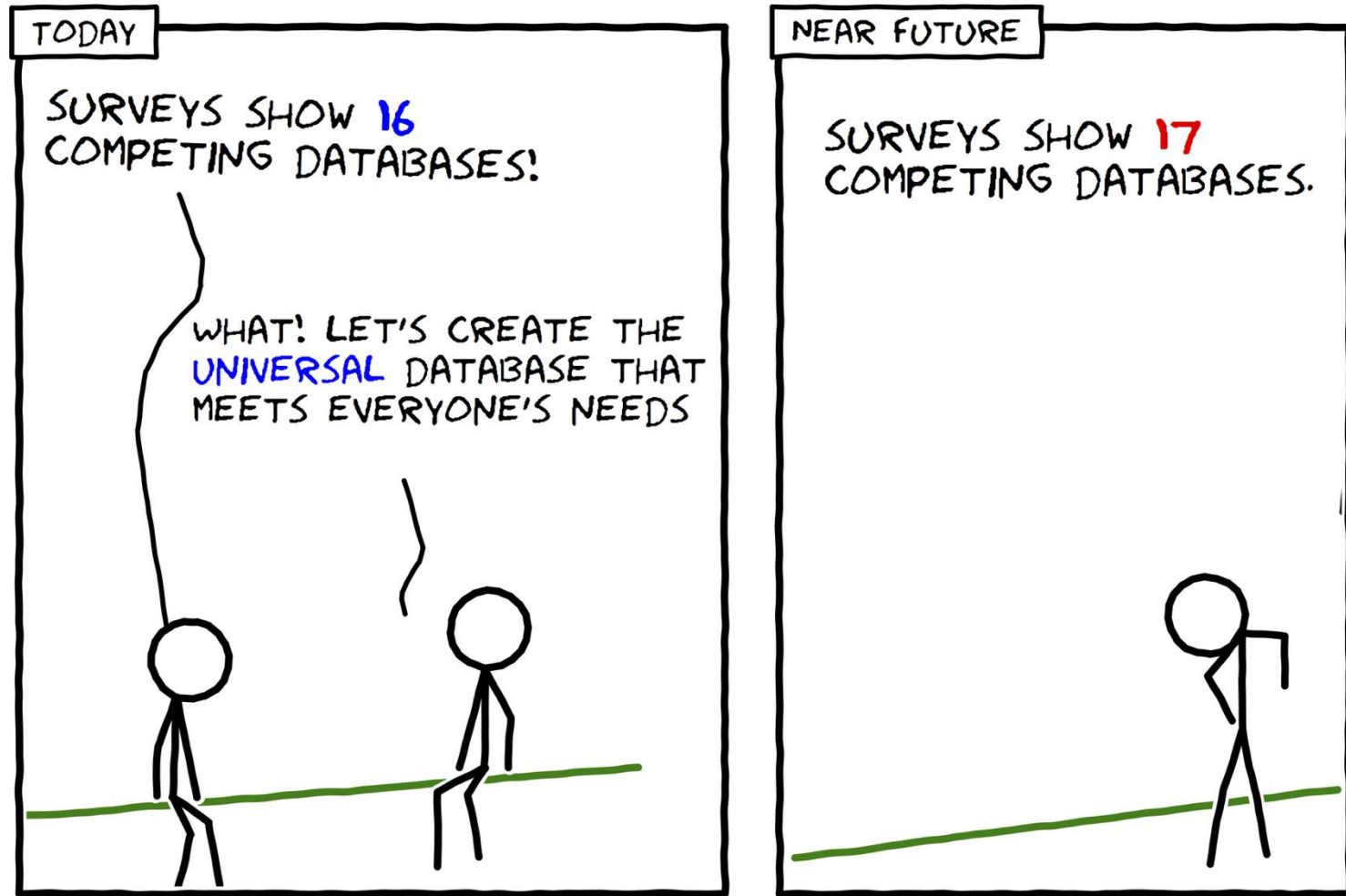


```
# Load modules
from matminer.datasets import get_available_datasets
from matminer.datasets import load_dataset

# See what's available
get_available_datasets()

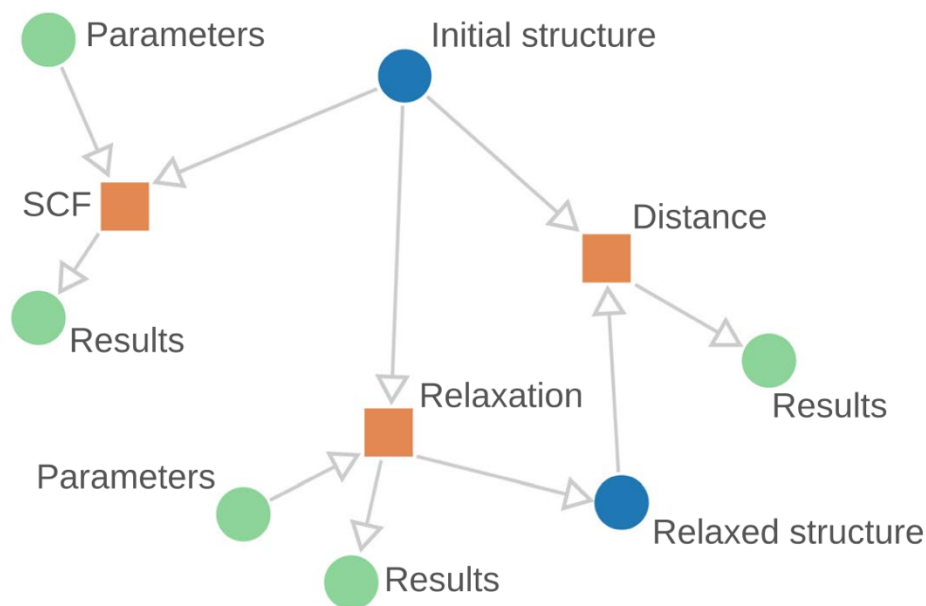
# Load dataset
df = load_dataset("citric_acid_thermal_conductivity")
```

Structure and Property Databases

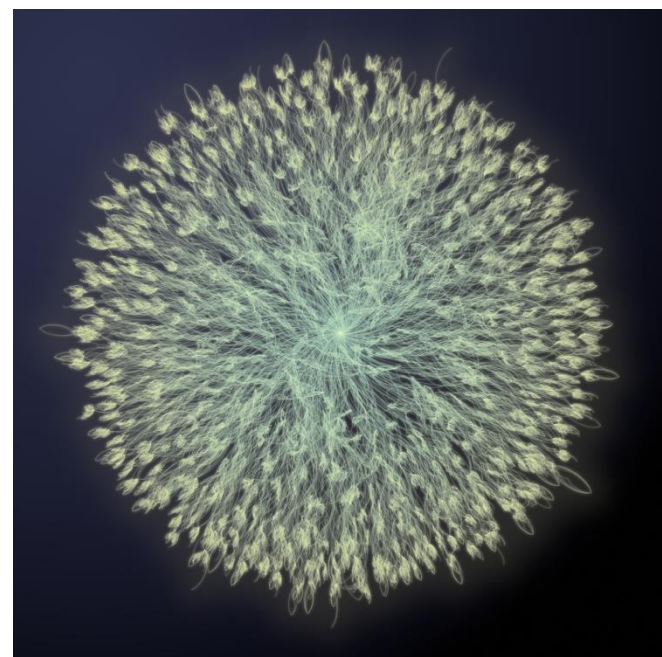


Data Provenance

Projects can combine data from many sources.
Provenance graphs are one way to link them



Connections between **structures**,
calculations, and **data**



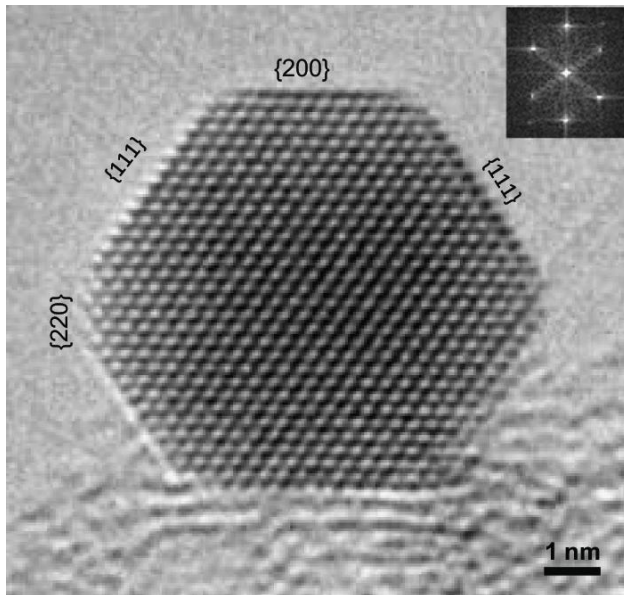
Graph for a project on 324
covalent organic frameworks

Image Data

Images are widely used in materials science.
The building blocks are pixels (e.g. 128×128)

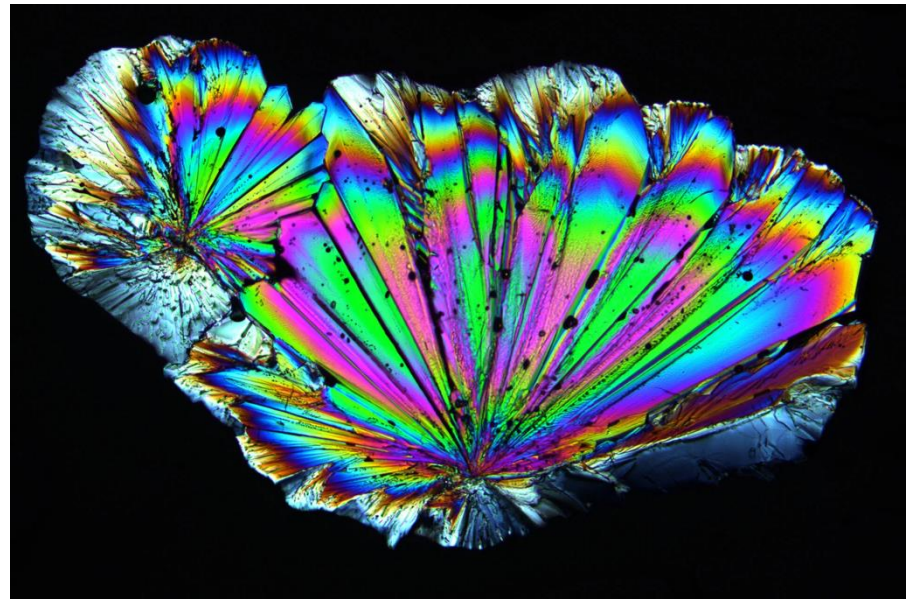
Greyscale

Pixel $\in [0,255]$



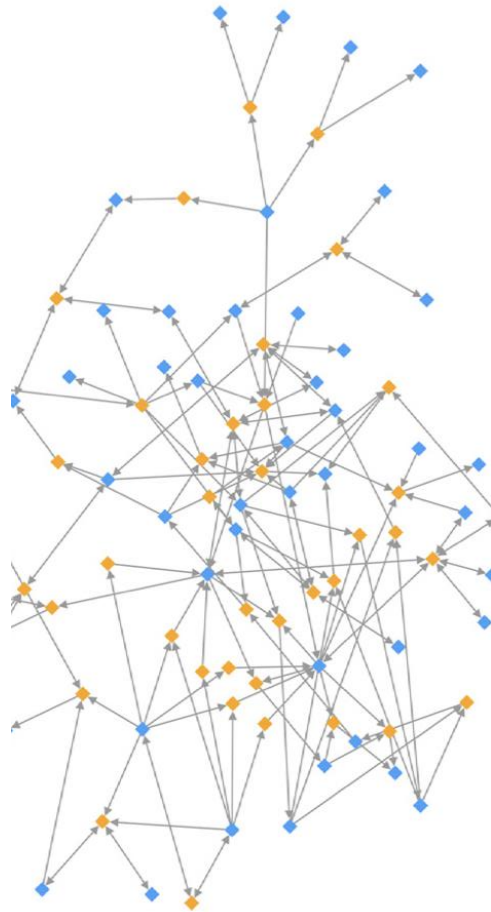
Colour

$P_R, P_G, P_B \in [0,255]$

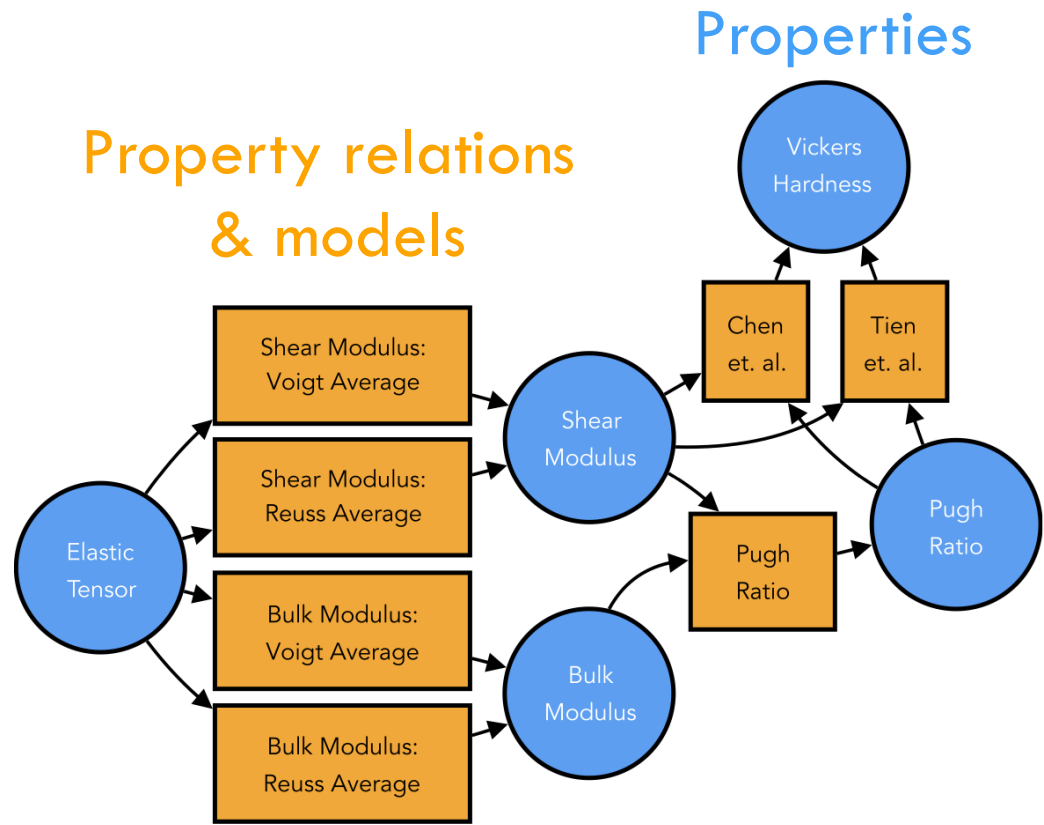


Knowledge Graphs

Structured representation of knowledge to model properties and their interrelations in a graph format



Property relations & models



Class Outcomes

1. Describe the importance of materials data for research and development
2. Demonstrate an understanding of the types of data that are shared in the materials community
3. Perform simple queries using Python APIs

Activity:

Data-driven thermoelectrics
